

Reconstruction of the initial functions from light cone traces of solutions of the fractional wave equation

Dabin Park
(working with S. Moon)

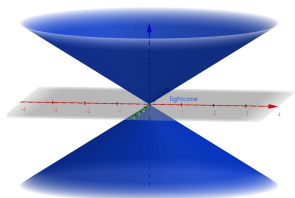
Kyungpook National University

YNMS
June 24, 2022

Introduction

light cone

- (a) The path in all directions through which the light can travel along space-time.
- (b) Satisfies the wave equation because light is a wave.
- (c) Can express light cone as $t = \pm|\mathbf{x}|$ and here we deal with $t = |\mathbf{x}|$ case.



time axis : one second
spatial axes : 300,000 km

Figure: light cone.

- The fractional wave equation

$$\begin{aligned} D_t^\alpha p(\mathbf{x}, t; \alpha) &= -(-\Delta_{\mathbf{x}})^{\frac{\alpha}{2}} p(\mathbf{x}, t; \alpha), \\ (\mathbf{x}, t) &= (x_1, x_2, t) \in \mathbb{R}^2 \times (0, \infty), \quad 1 < \alpha \leq 2 \end{aligned} \quad (1)$$

- The Riesz space-fractional derivative of order α

$$\mathcal{F}(-(-\Delta_{\mathbf{x}})^{\frac{\alpha}{2}} f)(\boldsymbol{\xi}) := -|\boldsymbol{\xi}|^\alpha (\mathcal{F}f)(\boldsymbol{\xi})$$

where $\mathcal{F}$: the Fourier transform

- The Caputo time-fractional derivative of order α

$$(D^\alpha h)(t) := (I^{m-\alpha} h^{(m)})(t), \quad m-1 < \alpha \leq m, \quad m \in \mathbb{N},$$

where I^α : the Riemann-Liouville fractional integral

$$(I^\alpha h)(t) := \begin{cases} \frac{1}{\Gamma(\alpha)} \int_0^t (t-\tau)^{\alpha-1} h(\tau) d\tau, & \text{if } \alpha > 0, \\ h(t), & \text{if } \alpha = 0 \end{cases}$$

Suppose $f \in C_c^\infty(\mathbb{R}^2)$.

Solution of
$$\begin{aligned} D_t^\alpha p(\mathbf{x}, t; \alpha) &= -(-\Delta_{\mathbf{x}})^{\frac{\alpha}{2}} p(\mathbf{x}, t; \alpha) \\ p(\mathbf{x}, 0; \alpha) &= f(\mathbf{x}), \text{ and } D_t p(\mathbf{x}, t; \alpha)|_{t=0} = 0 \end{aligned} :$$

$$p(\mathbf{x}, t; \alpha) = \frac{1}{(2\pi)^2} \int_{\mathbb{R}^2} e^{i\xi \cdot \mathbf{x}} E_\alpha(-|\xi|^\alpha t^\alpha) \mathcal{F}f(\xi) d\xi, \quad (2)$$

where E_α : the Mittag-Leffler function

$$E_\alpha(z) = \sum_{k=0}^{\infty} \frac{z^k}{\Gamma(1 + \alpha k)}, \quad \operatorname{Re}(\alpha) > 0, \quad z \in \mathbb{C}. \quad (3)$$

If $\alpha = 2$

$$\begin{aligned}\partial_t^2 p(\mathbf{x}, t) &= \Delta_{\mathbf{x}} p(\mathbf{x}, t) \\ p(\mathbf{x}, 0) &= f(\mathbf{x}), \text{ and } D_t p(\mathbf{x}, t)|_{t=0} = 0\end{aligned}$$

Since $E_2(-z^2) = \cos(z)$,

$$p(\mathbf{x}, t) = \frac{1}{(2\pi)^2} \int_{\mathbb{R}^2} e^{i\xi \cdot \mathbf{x}} \cos(|\xi|t) \mathcal{F}f(\xi) d\xi,$$

$$p(\mathbf{x}, t; \alpha) = \frac{1}{(2\pi)^2} \int_{\mathbb{R}^2} e^{i\xi \cdot \mathbf{x}} E_\alpha(-|\xi|^\alpha t^\alpha) \mathcal{F}f(\xi) d\xi.$$

- Define the linear operator $\mathcal{W}_\alpha$

$$\mathcal{W}_\alpha f(\mathbf{x}) = p(\mathbf{x}, |\mathbf{x}|; \alpha). \quad (4)$$

Goal : reconstruction $f(\mathbf{x})$ from $p(\mathbf{x}, |\mathbf{x}|; \alpha)$

When the spatial dimension is odd, Rakesh and Yuan have already provide the inverting the operator trace the solution on the light cone from the initial function [4].

Strategy

$$\mathcal{W}_\alpha f \longmapsto (\mathcal{W}_\alpha f)_m \longmapsto (\mathcal{F}f)_m \longmapsto \mathcal{F}f \longmapsto f$$

where $(\mathcal{W}_\alpha f)_m$: Fourier coefficient of $\mathcal{W}_\alpha f$

Fourier series of $\mathcal{W}_\alpha f$ and $\mathcal{F}f$

$$\mathcal{W}_\alpha f(r_{\mathbf{x}}\boldsymbol{\theta}_{\mathbf{x}}) = \sum_{m=-\infty}^{\infty} (\mathcal{W}_\alpha f)_m(r_{\mathbf{x}}) Y_m(\boldsymbol{\theta}_{\mathbf{x}}) \quad \text{and} \quad (5)$$

$$\mathcal{F}f(r_{\boldsymbol{\xi}}\boldsymbol{\theta}_{\boldsymbol{\xi}}) = \sum_{m=-\infty}^{\infty} (\mathcal{F}f)_m(r_{\boldsymbol{\xi}}) Y_m(\boldsymbol{\theta}_{\boldsymbol{\xi}}), \quad (6)$$

Here $Y_m(\theta) = \frac{1}{\sqrt{2\pi}} e^{im\theta}$,

$(r_{\mathbf{x}}, \boldsymbol{\theta}_{\mathbf{x}}) \in [0, \infty) \times \mathbb{S}^1$ with $\mathbf{x} = r_{\mathbf{x}}\boldsymbol{\theta}_{\mathbf{x}}$, where $\boldsymbol{\theta}_{\mathbf{x}} = (\cos \theta_{\mathbf{x}}, \sin \theta_{\mathbf{x}})$,
 $(r_{\boldsymbol{\xi}}, \boldsymbol{\theta}_{\boldsymbol{\xi}}) \in [0, \infty) \times \mathbb{S}^1$ with $\boldsymbol{\xi} = r_{\boldsymbol{\xi}}\boldsymbol{\theta}_{\boldsymbol{\xi}}$, where $\boldsymbol{\theta}_{\boldsymbol{\xi}} = (\cos \theta_{\boldsymbol{\xi}}, \sin \theta_{\boldsymbol{\xi}})$.

Mellin transform

Let $f(x)$ be a locally integrable function on $(0, \infty)$

$$\mathcal{M}f(s) = \int_0^{\infty} x^{s-1} f(x) dx, \quad s \in \mathbb{C}. \quad (7)$$

If

$$f(x) = \begin{cases} O(x^{-a-\epsilon}), & \text{as } x \rightarrow 0^+, \\ O(x^{-b+\epsilon}), & \text{as } x \rightarrow +\infty, \end{cases} \quad \epsilon > 0, \quad a < b \quad (8)$$

then the integral (7) converges absolutely and defines an analytic function in the strip $a < \operatorname{Re}(s) < b$.

$$f(x) = \frac{1}{2\pi i} \int_{c-i\infty}^{c+i\infty} t^{-s} \mathcal{M}f(s) ds, \quad a < c < b \quad (9)$$

Property of Mellin transform :

$$\begin{aligned} \mathcal{M}(f \times g) &= \mathcal{M}f \cdot \mathcal{M}g, \\ f \times g(t) &= \int_0^{\infty} f(\tau) g\left(\frac{t}{\tau}\right) \frac{d\tau}{\tau}. \end{aligned}$$

Main results

Strategy

$$\mathcal{W}_\alpha f \longmapsto (\mathcal{W}_\alpha f)_m \longmapsto (\mathcal{F}f)_m \longmapsto \mathcal{F}f \longmapsto f$$

- The relation between $(\mathcal{F}f)_m$ and $(\mathcal{W}_\alpha f)_m$

Theorem 1

For $f \in C_c^\infty(\mathbb{R}^2)$, $(\mathcal{W}_\alpha f)_m(r) = \frac{i^m}{2\pi} \int_0^\infty E_\alpha(-\rho_\xi^\alpha r^\alpha) (\mathcal{F}f)_m(\rho_\xi) \rho_\xi J_m(\rho_\xi r) d\rho_\xi$,

where $J_\nu(\cdot)$: Bessel function (of the first kind) of order ν .

- To use Mellin transform

$$r^{-2}(\mathcal{W}_\alpha f)_m(r^{-1}) = \frac{(-1)^{|m|} i^m}{2\pi} (\mathcal{F}f)_m \times E(r; \alpha, m), \quad (10)$$

where $E(r; \alpha, m) = E_\alpha(-r^{-\alpha}) J_{|m|}(r^{-1}) r^{-2}$.

Theorem 2

Let $f \in C_c^\infty(\mathbb{R}^2)$. For $\frac{3}{2} + \epsilon < c < 2 + |m| - \epsilon$ with $\epsilon < \frac{1}{4}$, we have

$$(\mathcal{F}f)_m(r) = i^{m-1} \int_{c-i\infty}^{c+i\infty} r^{-s} \frac{\mathcal{M}[(\cdot)^{-2}(\mathcal{W}_\alpha f)_m(\cdot^{-1})](s)}{\mathcal{M}(E)(s; \alpha, m)} ds.$$

Theorem 1

For $f \in C_c^\infty(\mathbb{R}^2)$. $(\mathcal{W}_\alpha f)_m(r) = \frac{i^m}{2\pi} \int_0^\infty E_\alpha(-\rho_\xi^\alpha r^\alpha) (\mathcal{F}f)_m(\rho_\xi) \rho_\xi J_m(\rho_\xi r) d\rho_\xi$,

where $J_\nu(\cdot)$: Bessel function (of the first kind) of order ν .

Proof

$$\begin{aligned}\mathcal{W}_\alpha f(\mathbf{x}) &= \frac{1}{(2\pi)^2} \int_{\mathbb{S}^1} \int_0^\infty e^{i\rho_\xi \theta_\xi \cdot \mathbf{x}} E_\alpha(-\rho_\xi^\alpha |\mathbf{x}|^\alpha) \mathcal{F}f(\rho_\xi \theta_\xi) \rho_\xi d\rho_\xi dS(\theta_\xi) \\ &= \frac{1}{(2\pi)^2} \sum_{m=-\infty}^\infty \int_0^\infty E_\alpha(-\rho_\xi^\alpha |\mathbf{x}|^\alpha) (\mathcal{F}f)_m(\rho_\xi) \rho_\xi \left(\int_{\mathbb{S}^1} e^{i\rho_\xi \theta_\xi \cdot \mathbf{x}} Y_m(\theta_\xi) dS(\theta_\xi) \right) d\rho_\xi \\ &= \sum_{m=-\infty}^\infty \frac{i^m}{2\pi} \int_0^\infty E_\alpha(-\rho_\xi^\alpha |\mathbf{x}|^\alpha) (\mathcal{F}f)_m(\rho_\xi) \rho_\xi J_m(\rho_\xi |\mathbf{x}|) d\rho_\xi Y_m\left(\frac{\mathbf{x}}{|\mathbf{x}|}\right),\end{aligned}$$

The Funk-Hecke theorem in [2, p 198] :

$$\int_{\mathbb{S}^1} e^{i\rho_\xi |\mathbf{x}| \theta_\xi \cdot \frac{\mathbf{x}}{|\mathbf{x}|}} Y_m(\theta_\xi) dS(\theta_\xi) = (2\pi) i^m J_m(\rho_\xi |\mathbf{x}|) Y_m\left(\frac{\mathbf{x}}{|\mathbf{x}|}\right).$$

Well - defined

- to Use the mellin transform

$$r^{-2}(\mathcal{W}_\alpha f)_m(r^{-1}) = \frac{(-1)^{|m|} i^m}{2\pi} (\mathcal{F}f)_m \times E(r; \alpha, m),$$

where $E(r; \alpha, m) = E_\alpha(-r^{-\alpha})J_{|m|}(r^{-1})r^{-2}$, beacuse $J_m = (-1)^{|m|}J_{|m|}$.

Since $f \in C_c^\infty(\mathbb{R}^2)$, $\mathcal{F}f$ decays quickly and so $(\mathcal{F}f)_m$ still decays quickly. $\mathcal{M}((\mathcal{F}f)_m)(s)$ and inversion formula for $\mathcal{M}((\mathcal{F}f)_m)(s)$ are well defined for $0 + \epsilon < \operatorname{Re}(s)$.

Lemma

If $E(r; \alpha, m) = E_\alpha(-r^{-\alpha})J_{|m|}(r^{-1})r^{-2}$, then we have

$$E(r; \alpha, m) = \begin{cases} O(r^{-(\frac{3}{2})}), & \text{as } r \rightarrow 0^+, \\ O(r^{-(2+|m|)}), & \text{as } r \rightarrow +\infty, \end{cases}$$

so $\mathcal{M}(E)(s; \alpha, m)$ and inversion formula for $\mathcal{M}(E)(s; \alpha, m)$ are well defined for $\frac{3}{2} + \epsilon < \operatorname{Re}(s) < 2 + |m| - \epsilon$ with $\epsilon < \frac{1}{4}$.

Lemma

If $E(r; \alpha, m) = E_\alpha(-r^{-\alpha})J_{|m|}(r^{-1})r^{-2}$, then we have

$$E(r; \alpha, m) = \begin{cases} O(r^{-(\frac{3}{2})}), & \text{as } r \rightarrow 0^+, \\ O(r^{-(2+|m|)}), & \text{as } r \rightarrow +\infty, \end{cases}$$

so $\mathcal{M}(E)(s; \alpha, m)$ and inversion formula for $\mathcal{M}(E)(s; \alpha, m)$ are well defined for $\frac{3}{2} + \epsilon < \operatorname{Re}(s) < 2 + |m| - \epsilon$ with $\epsilon < \frac{1}{4}$.

Proof

$E(r; \alpha, m) = O(r^{-(\frac{3}{2})})$ as $r \rightarrow 0^+$.

- $E_\alpha(-r^{-\alpha}) = O(r^0)$ as $r \rightarrow 0^+$
- $J_\nu(x) \approx x^{-\frac{1}{2}} \cos\left(x - \frac{(2\nu+1)\pi}{4}\right)$ for large x , so $J_{|m|}(r^{-1}) = O(r^{\frac{1}{2}})$ as $r \rightarrow 0^+$

$E(r; \alpha, m) = O(r^{-(2+|m|)})$ as $r \rightarrow +\infty$.

- $E_\alpha(-r^{-\alpha})$ has finite value as $r \rightarrow +\infty$, so $E_\alpha(-r^{-\alpha}) = O(r^0)$ as $r \rightarrow +\infty$
- $J_\nu(x) \approx \frac{x^\nu}{2^\nu \Gamma(\nu+1)}$ for x near 0, so $J_{|m|}(r^{-1}) = O(r^{-|m|})$ as $r \rightarrow +\infty$

Theorem 2

Let $f \in C_c^\infty(\mathbb{R}^2)$. For $\frac{3}{2} + \epsilon < c < 2 + |m| - \epsilon$ with $\epsilon < \frac{1}{4}$, we have

$$(\mathcal{F}f)_m(r) = i^{m-1} \int_{c-i\infty}^{c+i\infty} r^{-s} \frac{\mathcal{M}[(\cdot)^{-2}(\mathcal{W}_\alpha f)_m(\cdot^{-1})](s)}{\mathcal{M}(E)(s; \alpha, m)} ds.$$

Proof

$$\begin{aligned} \mathcal{M}[(\cdot)^{-2}(\mathcal{W}_\alpha f)_m(\cdot^{-1})](s) &= (-1)^{|m|} i^m (2\pi)^{-1} \mathcal{M}[(\mathcal{F}f)_m \times E](s; \alpha, m) \\ &= (-1)^{|m|} i^m (2\pi)^{-1} \mathcal{M}((\mathcal{F}f)_m)(s) \cdot \mathcal{M}(E)(s; \alpha, m). \end{aligned}$$

Taking the Mellin inverse formula (9) of

$$\mathcal{M}((\mathcal{F}f)_m)(s) = \frac{i^m (2\pi) \mathcal{M}[(\cdot)^{-2}(\mathcal{W}_\alpha f)_m(\cdot^{-1})](s)}{\mathcal{M}(E)(s; \alpha, m)},$$

we have

$$(\mathcal{F}f)_m(t) = i^{m-1} \int_{c-i\infty}^{c+i\infty} t^{-s} \frac{\mathcal{M}[(\cdot)^{-2}(\mathcal{W}_\alpha f)_m(\cdot^{-1})](s)}{\mathcal{M}(E)(s; \alpha, m)} ds.$$

Case $n = 3$

$Y_{lk}(\theta, \varphi)$ the spherical harmonics, which form an orthonormal basis in $L^2(\mathbb{S}^2)$

$$Y_{lk}(\theta, \varphi) = \sqrt{\frac{2l+1}{4\pi} \frac{(l-|k|)!}{(l+|k|)!}} P_l^{|k|}(\cos \theta) e^{ik\varphi}, \quad k = -l, \dots, l, \quad l = 0, 1, 2, \dots$$

where θ : polar angle

φ : azimuthal angle

P_l^k : associated Legendre functions

$$\left((1-t^2)(P_l^k)''(t) - 2t(P_l^k)'(t) + \left\{ l(l+1) - \frac{k^2}{1-t^2} \right\} P_l^k(t) \right) = 0$$

In three dimensions, f can be reconstructed from $\mathcal{W}_\alpha f$ in the same way as reconstructed in two dimensions.

In all dimensions, we can reconstruct f from $\mathcal{W}_\alpha f$.

Reference



R. Gorenflo, A. A. Kilbas, F. Mainardi and S.V. Rogosin
Mittag-Leffler Functions, Related Topics and Applications.
Springer Monographs Mathematics, Springer, Heidelberg, 2014.



F. Natterer.
The Mathematics of Computerized Tomography.
Classics in Applied Mathematics. Society of Industrial and Applied Mathematics, Philadelphia, 2001.



D. Park and S. Moon
Reconstruction of the initial functions from the solutions of the
fractional wave equation on the light cone trace



Rakesh and T Yuan.
Recovering initial values from light cone traces of solutions of the
wave equation.
Inverse Problems, 34(7):075003, 2018.

Thank you.